C Programming — Pointers

Theory Questions & Programming Problems | Intern Training Session

Section A: Theory Questions

1 What is a pointer in C? Why do we use pointers?
2 What is the difference between a normal variable and a pointer variable?
3 How do you declare and initialize a pointer?
4 What is the meaning of the & (address-of) and * (dereference) operators?
5 What is a NULL pointer? Why is it used?
6 What is a wild/dangling pointer? How can it cause problems?
7 What is the size of a pointer variable? Does it depend on the data type it points to?
8 What is a void pointer? Where is it used?
9 What is pointer arithmetic? Give examples of valid pointer operations.
10 What happens when you add 1 to a pointer (pointer increment)?
11 What is the relationship between arrays and pointers?
12 What is a pointer to a pointer (double pointer)? Give an example use case.
13 What is the difference between call by value and call by reference using pointers?
14 What is a function pointer? Why is it useful?
15 What is the difference between *ptr++ and (*ptr)++?
16 What is dynamic memory allocation? Name the functions used for it (malloc, calloc, realloc, free).
17 What is the difference between malloc() and calloc()?
18 What is a memory leak, and how do pointers cause it?
19 What is the difference between a constant pointer and a pointer to a constant?
20 What is pointer-to-array vs array-of-pointers? How do they differ?

Section B: Programming Problems

Basic

1 Write a program to declare a pointer and print the value and address of a variable using it.
2 Write a program to swap two numbers using pointers (call by reference).
3 Write a program to add two numbers using pointers.
4 Write a program to find the size of different data types using pointers and sizeof.
5 Write a program to demonstrate pointer arithmetic (increment/decrement) on an integer pointer.

Pointers with Arrays

6 Write a program to access array elements using a pointer instead of array indexing.

7 Write a program to find the sum of array elements using pointers.

8 Write a program to reverse an array using pointers.

9 Write a program to find the largest element in an array using pointers.

10 Write a program to copy one array into another using pointers.

Pointers with Strings

11 Write a program to find the length of a string using pointers (without strlen).

12 Write a program to copy a string using pointers (without strcpy).

13 Write a program to reverse a string using pointers.

14 Write a program to compare two strings using pointers (without strcmp).

15 Write a program to concatenate two strings using pointers (without strcat).

Pointers with Functions

16 Write a program to swap two numbers by passing pointers to a function.

17 Write a program to pass an array to a function using a pointer and modify its elements.

18 Write a program demonstrating a function pointer that calls add() or subtract() based on user input.

19 Write a program to return multiple values from a function using pointers.

Double Pointers / Dynamic Memory

20 Write a program to demonstrate a pointer to a pointer (double pointer) and print values at each level.

21 Write a program to dynamically allocate memory for an array using malloc() and free it.

22 Write a program to dynamically allocate memory for an array using calloc() and compare with malloc().

23 Write a program to resize a dynamically allocated array using realloc().

24 Write a program to create a 2D array dynamically using pointer-to-pointer and malloc().

Advanced

25 Write a program to implement a simple linked list using structures and pointers (insert and display).

26 Write a program to demonstrate the difference between a constant pointer and a pointer to a constant.

27 Write a program to sort an array using pointers and the bubble sort technique.